Hazem Nabil Alieldin

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OBJECTIVES

AI & Data Science Engineer with solid internship experience in machine learning, predictive analytics, Agentic AI, LLM apps with prompt engineering and data visualization. Proficient in Python, TensorFlow, Scikit-learn, and SQL with strong foundations in data preprocessing, model development, and cloud tools. Passionate about extracting actionable insights, automating data pipelines, and building scalable AI solutions.

PROFESSIONAL EXPERIENCE

- **Information technology institute – ITI**
  Building LLM apps with prompt engineering and RAG systems, January 2026 - February 2026, online
  - Engineered Retrieval-Augmented Generation (RAG) pipelines by integrating LLMs with vector databases (e.g., Pinecone, Milvus, or Weaviate) to enable semantic search over private datasets.
  - Architected end-to-end LLM workflows using frameworks like LangChain or LlamaIndex to automate complex business logic and decision-making processes.
  - Optimized model performance through advanced prompt engineering, fine-tuning, and evaluation frameworks (e.g., RAGAS) to ensure factual accuracy and reduce hallucinations.

- **National Telecommunication Institute - NTI**
  HUAWEI Big Data, July 2025 - August 2025, Remote
  - Analyzed large datasets and provided actionable insights using SQL and Power BI.
  - Created interactive dashboards to visualize business trends Authored technical documentation for model
  - Engaged with Huawei Big Data tools and ecosystem to manage large-scale data processing tasks

- **Mindset**
  AI & Data Science Trainee and Mentor, October 2024 - June 2025, Hybrid
  - Mentored and guided students in Artificial Intelligence projects, covering model development, data analysis, and deployment.
  - Evaluated and graded student assignments, ensuring accuracy, quality, and adherence to learning objectives.
  - Conducted student interviews and assessed technical and soft skills for program placement.

- **Electronics research institute**
  HCIA-AI Intern, September 2024 - November 2024, Remote
  - Trained large language models (LLMs) to solve complex mathematical problems with high accuracy.
  - Collaborated with cross-functional teams to design and implement data-driven AI solutions.
  - Optimized model performance through iterative testing and refinement, enhancing solution reliability.

- **National Telecommunication Institute - NTI**
  AI for Business, July 2024 - July 2024, On-site
  - Applied Artificial Intelligence techniques to real-world business scenarios, including data-driven decision-making and process automation.
  - Analyzed datasets to extract insights and support strategic planning.
  - Developed predictive models to optimize operations and enhance customer experience.

EDUCATION

Egypt Japan University of Science & Technology
***Bachelor, Computer Science and Artificial Intelligence**, 2022 - 2026*
- CGPA - 3.4

CERTIFICATIONS

- Deep Learning Fundamentals, NVIDIA
- Business Startup Camp Simulation - E-JUST
- Innovation and Entrepreneurship, ITIDA
- Python Programming, Mahara Tech – ITIMooca
- Data analysis using Excel – Eng/Alaa Essam
- AI and Machine Learning – Mindset
- ECPC competitive programing 2024 - AAST

PROJECTS

View All Projects

- **Audio Classification** **July 2025**
  - Developed an audio classification system to categorize environmental sounds using machine learning.
  - Processed UrbanSound8K dataset, extracted features (MFCCs), and trained CNN models in Python..
  - Achieved high classification accuracy using TensorFlow and Librosa for preprocessing and model tuning.
- **AI Chatbot (NLP-based),** **April 2025**
  - Built an AI chatbot using Natural Language Processing for user interaction and task automation.
  - Trained models using Rasa and Transformer-based architectures to handle intents and dialogue management.
  - Integrated with a web interface for live testing and feedback collection.
- **Arabic Handwriting System** **May 2025**
  - Designed and implemented a deep learning-based system to recognize Arabic handwritten text.
  - Utilized Convolutional Neural Networks (CNNs) with TensorFlow and Keras for image classification.
  - Preprocessed datasets using OpenCV and NumPy to enhance recognition accuracy.
- **GAN-based Image Generation** **June 2025**
  - Developed a Generative Adversarial Network (GAN) to generate synthetic images from random noise input.
  - Built and trained Generator and Discriminator models using TensorFlow and Keras.
  - Applied Deep Convolutional GAN (DCGAN) architecture on datasets like MNIST and Fashion-MNIST.
- **Human activity detection** **December 2024 - January 2025**
  - Implemented a human activity recognition system using sensor data (e.g., accelerometer, gyroscope) to classify user movements.
  - Applied LSTM and Random Forest models to sequence data for accurate multi-class classification.
  - Used Scikit-learn and TensorFlow for training and evaluation.

SKILLS

- **Programming & Tools:** Python, SQL, MySQL, Git, GitHub, Jupyter Notebook, Google Colab, C++, assembly, Data structure, Algorithm
- **Machine Learning:** Scikit-Learn, XGBoost, GridSearchCV, Feature Engineering, MLFlow, Image Preprocessing.
- **Deep Learning:** TensorFlow, Keras, PyTorch, CNN, Neural Networks, Computer Vision, Yolo, Object Detection, Semantic Segmentation.
- **Data Analysis:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Data Preprocessing, Data Visualization.
- **Interpersonal Skills:** Management, Organization Skills, Problem Solving, Communication, Teamwork.
- **Languages:** English B2, Arabic Native, German Beginner

Volunteering

- **JLPT Exam organizer at EJSUT (Japanese language proficiency test)**
  Organizer **Jun 2025 - Present**
  - Guided participants in projects, promoting collaboration on science and technology challenges.
  - Fostered problem-solving skills, enabling teams to develop creative solutions for complex applications.
  - Facilitated interdisciplinary teamwork to advance project outcomes in exploration and technology.
- **ECPC** competitive programing
  Event Organizer, **March 2025**
  - Assisted in organizing a countrywide AI and Robotics programming competition with a capacity of 5,000, including 1,500 participants and 3,500 parents/spectators.